Randomized Algorithm

Pritha Banerjee

University of Calcutta

banerjee.pritha74@gmail.com

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Overview

1. Randomized Algorithm
2. Randomized QuickSort
3. Randomized Selection
4. Closest Pair
   - Using Divide and Conquer
   - Using Randomization
   - Design of Algorithm
   - Data structure for Subsquares
     - Hashing: A Randomization Implementation of Dictionary
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5. Books etc.

Deterministic Algorithm

- On input A, determinstic algorithm must produce a correct solution
- The number of computation steps must be same for different runs of algo with same input
- It may be difficult to formulate an algorithm with good runtime for sufficiently large input

How to get solutions quickly? Randomized Algorithm

Randomized Algorithm (1)

- On input A, randomized algorithm uses random choices to produce the correct solution
- It may go through different steps for different runs of algorithm with same input due to random choices.
- **Advantages:**
  - Randomized algorithms are often simple and easy to implement
  - Usually fast with high probability compared to deterministic algorithms.
  - Produces optimum output with very high probability
  - Malicious adversaries cannot generate any worst case input, that would slow down the randomized algorithm
- **Disadvantages:**
  - There is a finite probability of getting incorrect answer. But the probability can be made arbitrarily small by repeated application of randomness
  - Analysis of runtime and probability of getting correct answer is complex
  - Getting true random number is impossible and results depends on quality of random numbers.

Randomized Algorithm (2)

- Compute expected running time of randomized algorithm, as runtime is independent of any particular input.
- Inputs follow a probability distribution; thus, probabilistic analysis of expected running time

Find an a in input A where $n/2$ elements are a and rest b : Las Vegas Vs. Monte Carlo Algorithm

```
 findingA_LV(array A, n)
begin
    repeat
        Randomly select one element out of n elements.
    until 'a' is found
end

findingA_MC(array A, n, k)
begin
    i := 0
    repeat
        Randomly select one element out of n elements.
        i := i + 1
    until i = k or 'a' is found
end
```

Randomized Algorithm (3)

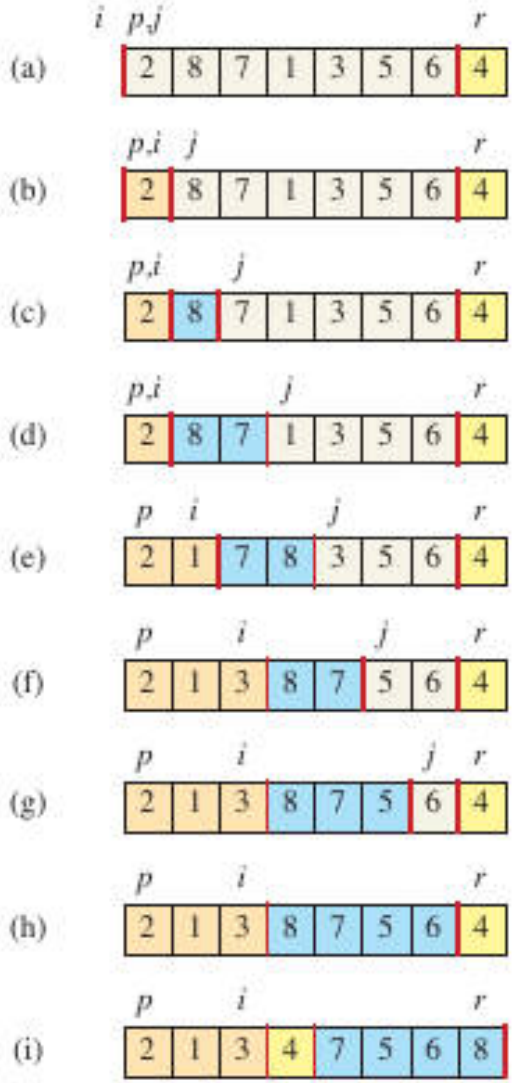
findingA(LV):

- Algo succeeds with probability 1, though number of iterations may vary.
- Expected number of iterations: $\lim_{n\to\infty} \sum_{i=1}^{n} \frac{i}{2^i} = 2$, ie, $\Theta(1)$

findingA(MC):

- if an a is found algo. succeeds, else it fails.
- After$k$ iterations, the probability of finding an 'a': $P\left[find\ a\right] = 1 - (\frac{1}{2})^k$
- The number of iterations is always less than or equal to k. k being constant the run time (expected and absolute) is $\Theta(1)$.

Quicksort




```
QUICKSORT(A, p, r)
if p < r
    // Partition the subarray around the pivot, which ends up in A[q].
    q = PARTITION(A, p, r)
    QUICKSORT(A, p, q − 1)   // recursively sort the low side
    QUICKSORT(A, q + 1, r)   // recursively sort the high side
```

```
PARTITION(A, p, r)
x = A[r]                                  // the pivot
i = p − 1                                 // highest index into the low side
for j = p to r − 1                        // process each element other than the pivot
    if A[j] ≤ x                           // does this element belong on the low side?
        i = i + 1                             // index of a new slot in the low side
        exchange A[i] with A[j]               // put this element there
exchange A[i + 1] with A[r]               // pivot goes just to the right of the low side
return i + 1                              // new index of the pivot
```

Randomized Quicksort

- Instead of always using $A[r]$ as pivot, select a randomly chosen element from $A[p..r]$

```
RANDOMIZED-QUICKSORT(A, p, r)
if p < r
    q = RANDOMIZED-PARTITION(A, p, r)
    RANDOMIZED-QUICKSORT(A, p, q − 1)
    RANDOMIZED-QUICKSORT(A, q + 1, r)
```

```
RANDOMIZED-PARTITION(A, p, r)
i = RANDOM(p, r)
exchange A[r] with A[i]
return PARTITION(A, p, r)
```

```
PARTITION(A, p, r)
x = A[r]
i = p − 1
for j = p to r − 1
    if A[j] ≤ x
        i = i + 1
        exchange A[i] with A[j]
exchange A[i + 1] with A[r]
return i + 1
```

Randomized Quicksort: Worst case time complexity (1)

- $T(n) = \max_{0 \le q \le n-1} (T(q) + T(n-q-1)) + \Theta(n)$
- solve by substitution method: Assume $T(n) \le cn^2$
- substitute the guess, ie,

$$\begin{aligned} T(n) &\le \max_{0 \le q \le n-1} (cq^2 + c(n-q-1)^2) + \Theta(n) \\ &= c \max_{0 \le q \le n-1} (q^2 + (n-q-1)^2) + \Theta(n) \end{aligned}$$

- $q^2 + (n-q-1)^2$ is maximum when $q = 0$ or $q = n-1$. For $q = 0, 1, \cdots, n-1$

$$\begin{aligned} q^2 + (n-q-1)^2) &= q^2 + (n-1)^2 - 2q(n-1) + q^2 \\ &= (n-1)^2 + 2q(q-(n-1)) \\ &\le (n-1)^2, \because q \le (n-1), 2q(q-(n-1)) \le 0 \end{aligned}$$

- Every term in the maximization is bounded by $(n-1)^2$

Randomized Quicksort: Worst case time complexity (2)

- $\max_{0 \le q \le n-1} (q^2 + (n-q-1)^2) \le (n-1)^2 = n^2 - 2n + 1$
- $T(n) \le cn^2 - c(2n-1) + \Theta(n)$
- since c can be taken large to dominate $\Theta(n)$, so $T(n) = O(n^2)$
- when partition is maximally unbalanced, $T(n) = \Omega(n^2)$, Thus, Worst case time complexity is $\Theta(n^2)$
- Randomized Quicksort is as bad as Quicksort for in the worst case.

Randomized Quicksort:Expected case time complexity (1)

- *PARTITION* is called n times throughout the quicksort, as during every partition a pivot is chosen and that pivot never appears later.
- Let X is the total number of comparisons in the entire quicksort algorithm (line 4 in Partition)
- Then the running time of QUICKSORT is $O(n + X)$
- Let A has elements $z_1, z_2, \cdots, z_n$, where z_i is the ith smallest element, so, define $Z_{ij} = \{z_i, z_{i+1}, \cdots, z_j\}$
- Need to know **when the algo. compares two elements z_i and z_j in the array and when it does not**
  - note that each pair of elements are compared only once
  - pivot is compared with all the elements only once, as that pivot does not participate later, once it is placed in correct position
- need probabilistic analysis: Let X_{ij} is indicator random variable

$$X_{ij} = 1, \text{when } z_i \text{ and } z_j \text{ is compared} \quad (1)$$

$$= 0, \text{otherwise} \quad (2)$$

Randomized Quicksort:Expected case time complexity (2)

- since each pair compared only once, total number of comparisons X is

$$X = \sum_{i=1}^{n-1} \sum_{j=i+1}^{n} X_{ij}$$

- taking expectation,

$$\begin{aligned} E(X) &= E\left[\sum_{i=1}^{n-1} \sum_{j=i+1}^{n} X_{ij}\right] \\ &= \sum_{i=1}^{n-1} \sum_{j=i+1}^{n} E\left[X_{ij}\right] \\ &= \sum_{i=1}^{n-1} \sum_{j=i+1}^{n} Pr\{z_i \text{ is compared to } z_j\} \end{aligned}$$

Randomized Quicksort:Expected case time complexity (3)

- Note 1): once a pivot x is chosen with $z_i < x < z_j$, then z_i and z_j are never compared any later time.
- Note 2): if z_i or z_j is pivot, then z_i or z_j is compared with any other item in Z_{ij}, except themselves
- what is the probability that z_i or z_j from Z_{ij} is the first element chosen as a pivot?
- any element of Z_{ij} is equally likely to be the first one chosen as pivot
- Pr { any elemnet is first one chosen as pivot from Z_{ij} } $=\frac{1}{(j-i+1)}$

$$
\begin{aligned}
Pr\{\ z_i \text{ is compared to } z_j\ \} &= Pr\{\ z_i \text{ or } z_j \text{ is first pivot chosen from } Z_{ij}\} \\
&= Pr\{\ z_i \text{ is first pivot chosen from } Z_{ij}\} \\
&+ Pr\{\ z_j \text{ is first pivot chosen from } Z_{ij}\} \\
&= \frac{1}{(j-i+1)} + \frac{1}{(j-i+1)}, \quad \text{mutually exclusive} \\
&= \frac{2}{(j-i+1)}
\end{aligned}
$$

Randomized Quicksort:Expected case time complexity (4)

$$
\begin{aligned}
E(X) &= \sum_{i=1}^{n-1} \sum_{j=i+1}^{n} \frac{2}{(j-i+1)}, \text{ harmonic series} \\
&= \sum_{i=1}^{n-1} \sum_{k=1}^{n-i} \frac{2}{(k+1)} \ , \ k = j - i \\
&< \sum_{i=1}^{n-1} \sum_{k=1}^{n-i} \frac{2}{(k)} \\
&= \sum_{i=1}^{n-1} O(\log n) = O(n \log n)
\end{aligned}
$$

Randomized Selection

Selection Problem: find k^{th} smallest element from $A[1..n]$

```
RANDOMIZED-SELECT(A, p, r, i)
if p == r
    return A[p]
q = RANDOMIZED-PARTITION(A, p, r)
k = q - p + 1
if i == k            // the pivot value is the answer
    return A[q]
elseif i < k
    return RANDOMIZED-SELECT(A, p, q - 1, i)
else return RANDOMIZED-SELECT(A, q + 1, r, i - k)
```

Randomized Selection : Time complexity (1)

- Worst case: $\Theta(n^2)$, if partitions always with the largest remaining element
- Expected running time: $O(n)$
- Run time $T(n)$ on input array $A[p..r]$ is a **random variable**; compute $E[T(n)]$
- Randomized partition is equally likely to return any element as pivot
- Thus, for each k such that $1 \leq k \leq n$, the subarray $A[p..q]$ has k elements $\leq$ pivot with probability $\frac{1}{n}$
- Let $X_k = 1$; $\{$the subarray $A[p..q]$ has exactly k elements $\}$, else $X_k = 0$
- $E[X_k] = P[X_k = 0].0 + P[X_k = 1].1 = \frac{1}{n}.1 = \frac{1}{n}$
- Compute an upper bound on time needed for recursive call on largest possible partition size $A[p..q-1]$ or $A[q+1..r]$

Randomized Selection : Time complexity (2)

- X_k has value 1 for exactly one value of k.

$$
\begin{aligned}
T(n) &\leq \sum_{k=1}^{n} X_k.(T(max(k-1, n-k)) + O(n)) \\
&= \sum_{k=1}^{n} X_k.T(max(k-1, n-k)) + \sum_{k=1}^{n} X_k.O(n) \\
&= \sum_{k=1}^{n} X_k.T(max(k-1, n-k)) + O(n); \ \because X_k = 1 \text{ for one } k
\end{aligned}
$$

Randomized Selection : Time complexity (3)

$$E[T(n)]$$

$$\leq E\left[\sum_{k=1}^{n} X_k.T(max(k-1, n-k)) + O(n)\right]$$

$$= \sum_{k=1}^{n} E[X_k.T(max(k-1, n-k))] + O(n); \text{ linearity of Expectation}$$

$$= \sum_{k=1}^{n} E[X_k].E[T(max(k-1, n-k))] + O(n); X_k \text{ and } T(max()) \text{ independent rv}$$

$$= \sum_{k=1}^{n} \frac{1}{n}.E[T(max(k-1, n-k))] + O(n); \text{ using } E[X_k] = \frac{1}{n}$$

What is $max(k-1, n-k)$**?**

$$max(k-1, n-k) = \begin{cases} k-1 & \text{if} k > \lceil n/2 \rceil \\ n-k & \text{if} k \leq \lceil n/2 \rceil \end{cases} \tag{3}$$

Randomized Selection : Time complexity (3)

Note that each term appears twice in $T(max(k-1, n-k))$, from $T(\lceil n/2 \rceil)$ to $T(n-1)$ for $k = 1 \cdots n$, thus,

$$= \frac{1}{n}(E[T(n-1)] + E[T(n-2)] \cdots + E[T(n/2+1)] +$$

$$E[T(n/2)+1] \cdots E[T(n-1)]) + O(n)$$

$$\text{Thus, } E[T(n)] \leq \frac{2}{n} \sum_{k=\lfloor n/2 \rfloor}^{n-1} E[T(k)] + O(n)$$

Use substitution to show $E[T(n)] = O(n)$, assume $E[T(n)] \leq cn$ and $T(n) = 1$ for n less that some constant $E[T(n)]$

$$\leq \frac{2}{n} \sum_{k=\lfloor n/2 \rfloor}^{n-1} ck + an$$

$$= \frac{2c}{n} \left(\sum_{k=1}^{n-1} k - \sum_{k=1}^{\lfloor n/2 \rfloor - 1} k \right) + an$$

Randomized Selection : Time complexity (4)

$$
\begin{aligned}
&= \frac{2c}{n}\left(\frac{(n-1)n}{2} - \frac{(\lfloor n/2 \rfloor - 1)\lfloor n/2 \rfloor}{2}\right) + an \\
&\leq \frac{2c}{n}\left(\frac{(n-1)n}{2} - \frac{(n/2-2)(n/2-1)}{2}\right) + an \\
&= \frac{2c}{n}\left(\frac{n^2-n}{2} - \frac{n^2/4 - 3n/2 + 2}{2}\right) + an \\
&= \frac{c}{n}\left(\frac{3n^2}{4} + \frac{n}{2} - 2\right) + an \\
&= c\left(\frac{3n}{4} + \frac{1}{2} - \frac{2}{n}\right) + an \\
&\leq \frac{3cn}{4} + \frac{c}{2} + an \\
&= cn - (\frac{cn}{4} - \frac{c}{2} - an)
\end{aligned}
$$

- To show for large n last expression is atmost cn, ie, $\frac{cn}{4} - \frac{c}{2} - an \geq 0$
- Add $c/2$ to both sides and factor out n to get $n(c/4 - a) \geq c/2$, ie, $(c/4 - a) > 0$, ie, $c > 4a$
- Thus, $n \geq \frac{c/2}{c/4-a} = \frac{2c}{c-4a}$
- For such n and c, $E[T(n)] = O(n)$

D & C Closest Pair Problem

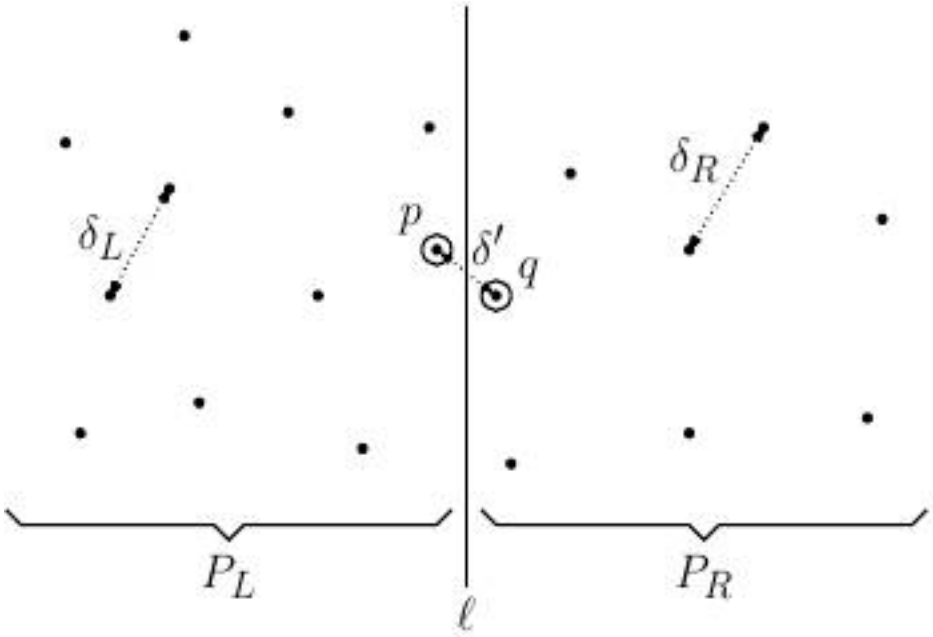
Problem: Given a set of points $P = \{p_1, p_2, \cdots, p_n\}$, where each p_i is (x_i, y_i), find the closest pair. Here closest pair is a pair of points whose Euclidean distance is minumum among all possible pairs.

- Given two points $p = (p_x, p_y)$ and $q = (q_x, q_y)$, their (Euclidean) distance is $||pq|| = \sqrt{(p_x - q_x)^2 + (p_y - q_y)^2}$.
- Brute force search: Takes $O(n^2)$ time, by computing the distance between each pair, and returning the smallest.
- Goal: $O(n \log n)$ time algorithm based on divide-and-conquer.

D & C Closest Pair Algorithm

- **Basis:** If $|P| \leq 3$, then just solve the problem by brute force in $O(1)$ time
- **Divide:** Otherwise, partition the points into two partitions P_L and P_R of roughly equal size (find median) based on their x-coordinates. Imagine a vertical line l that splits the points to P_L with points to the left of l and P_R, the points to the right of l.
- **Conquer:** Compute the closest pair within each of the subsets P_L and P_R each, by invoking the algorithm recursively. Let δ_L and δ_R be the closest pair distances in each case Let $\delta = min(\delta_L, \delta_R)$.
- **Combine:** Note that there may be two points p and q that are very close to one another but are on opposite sides of l. Need to find such closest pair (p, q) if exists with distance $\delta' < \delta$. If found, return $min(\delta, \delta')$ as the final result along with p, q being the closest.

D & C: Idea

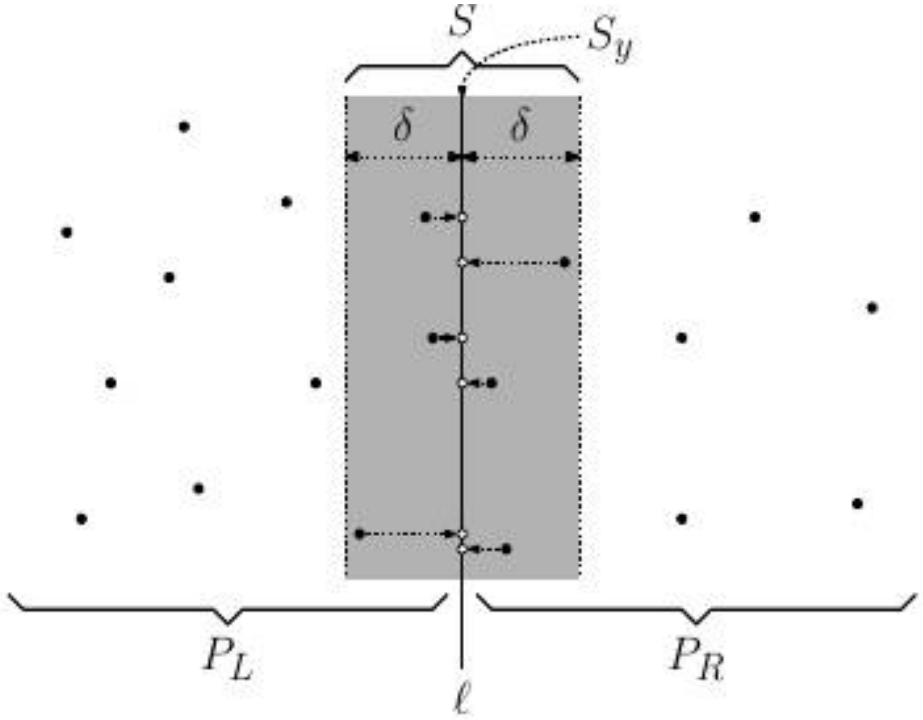


- **How to find median during each recursion (divide) ?** Expensive even if *Select* algorithm is used during every recursion. Hence **pre-sort** the x coordinate (P_x) and y coordinate (P_y) in $O(\log n)$ time only **once**. Use required subarray of P_x to find median during each recursion. In $O(n)$, partition the points into P_L and P_R
- If **combine** step can be done in $O(n)$ time, then time complexity is $T(n) = 2T(n/2) + O(n) = O(n \log n)$

How to combine? (1)

Need to compute the closest pair p and q, where $p \in P_L$ and $q \in P_R$.

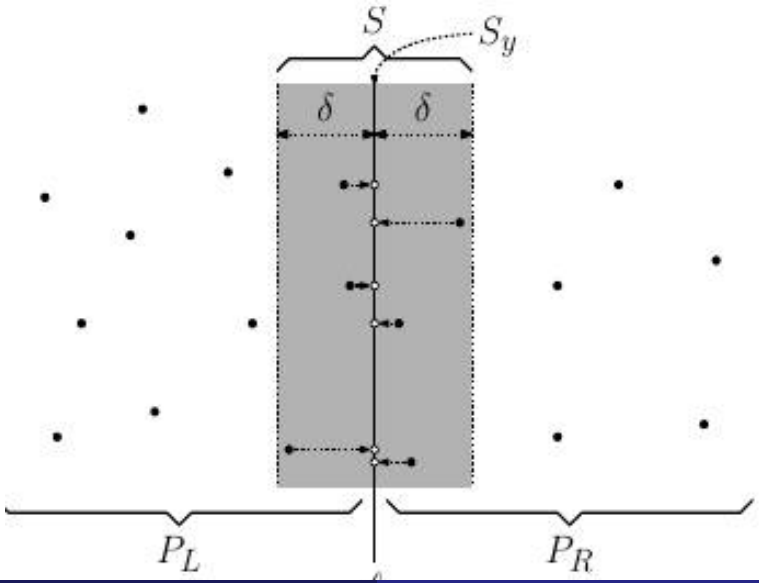
- We have the point set P , the x-coordinate of the vertical splitting line l, and the value of $\delta = min(\delta_L, \delta_R)$.
- Note that if such a pair of points p, q exists, then both points lie within distance δ of l, for otherwise the resulting distance would exceed δ.
- Let S denote this subset of P that lies within a vertical strip of width 2δ centered about l

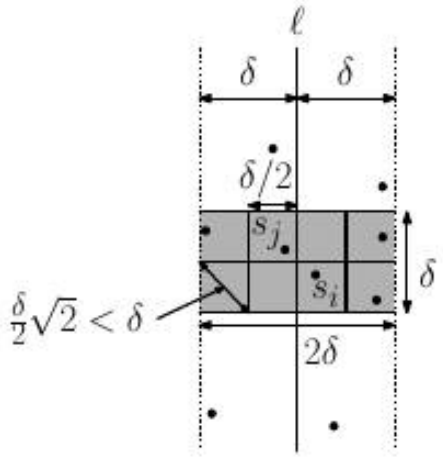


How to combine? (2)

How to find the closest pair within S?

- Let $Sy =< s_1, \cdots, s_m >$ are points of S sorted by their y-coordinates.
- If S_y contains two points that are within distance δ of each other, these two points must be within a **constant number of positions of each other** in the sorted array S_y.
- **Lemma:** Given any two points $s_i, s_j \in S_y$, if $||s_i s_j|| \leq \delta$, then $|j - i| \leq 7$.
- On the **sorted** S_y, compute distance δ' of a point with its 8 adjacent points in increasing order of S_y. If $\delta' < \delta$ update closest points and closest distance. Thus $O(n)$ time, if n points are in S_y





How to combine? (3)

Lemma: Given any two points $s_i, s_j \in S_y$, if $||s_i s_j|| \leq \delta$, then $|j - i| \leq 7$.
Proof:

- Suppose that $||s_i s_j|| \leq \delta$. Since they are in S they are each within distance δ of l.
- The y-coordinates of these two points can differ by at most δ. So they must both reside in a rectangle of width 2δ and height δ centered about l.
- Split this rectangle into **eight** identical squares each of side length $\delta/2$. A square of side length $\delta/2$ has a diagonal of length $\sqrt{(\delta/2)^2 + (\delta/2)^2} = \delta/\sqrt{2} < \delta$, and so no two points within such a square can be farther away than this.
- Since each square lies entirely on one side of l, no square can contain two or more points of P , since otherwise, these two points would contradict the fact that δ is the closest pair seen so far.
- Thus, there can be at most **eight** points of S in this rectangle, one for each square. Therefore, $|j - i| \leq 7$.

Computing S_y

- Sorting S_y explicitly is expensive, because we may have n points in S and we want $O(n)$ time for combine step
- Pre-sorting P_y helps.
- Compute S_y, enumerate the points of P_y, and each time we find a point that lies within the **strip**, copy it to the next position of array S_y . This runs in $O(n)$ time, and preserves the y-ordering of the points.

D & C Closest Pair Algo

```
closestPair(P = (Px, Py)) {
    n = |P|
    if (n <= 3) solve by brute force                // basis case
    else {
        Find the vertical line L through P's median // divide
        Split P into PL and PR (split Px and Py as well)
        dL = closestPair(PL)                        // conquer
        dR = closestPair(PR)
        d = min(dL, dR)
        for (i = 1 to n) {                          // create Sy
            if (Py[i] is within distance d of L) {
                append Py[i] to Sy
            }
        }
        d' = stripClosest(Sy)                       // closest in strip
        return min(d, d')                           // overall closest
    }
}

stripClosest(Sy) {                                  // closest in strip
    m = |Sy|
    d' = infinity
    for (i = 1 to m) {
        for (j = i+1 to min(m, i+7)) {              // search neighbors
            if (dist(Sy[i], Sy[j]) <= d') {
                d' = dist(Sy[i], Sy[j])             // new closest found
            }
        }
    }
    return d'
}
```

Closest Pair Problem and Randomization

Problem: Given a set of points $P = \{p_1, p_2, \cdots, p_n\}$, where each p_i is (x_i, y_i), find the closest pair.

- It uses randomization and dictionary data structure
- Algorithm runs in $O(n)$ expected time, plus $O(n)$ expected dictionary operations.
- Randomness happens as follows:
  - the way in which the algorithm processes the input points
  - when the dictionary is implemented using hashing
- Let $d(p_i, p_j)$ denotes the Euclidean distance between points p_i and p_j. Thus, find the pairs p_i and p_j such that $d(p_i, p_j)$ is minimized.
- Assume that the points are all in the unit square: $0 \leq x_i, y_i < 1$, $i = 1, 2, \cdots, n$

Designing the algorithm (1)

Overview:

- Consider the points in **random order**, and maintain a current value δ for closest pair
- When a new point p appears, look **in the vicinity** of p to see if any of the previously considered points are at distance less than δ from p
- **If not**, then closest pair has **not changed**, move to the next point in **random** order
- **Else**, there is a ponit closer to p, update δ and the closest pair

Stages of Algorithm:

- Assume **random order** of points are $p_1, p_2, \cdots, p_n$
- In each **stage**, closest pair remains constant
- **First stage** : $\delta = d(p_1, p_2)$
- During a **stage** points get added as long as there is **no change in closest pair**
- Stage terminates for a point p_i, when some $p_j, j < i$, $d(p_i, p_j) < \delta$
- Thus $\delta = \min_{j:j<i} d(p_i, p_j)$

Designing the algorithm (2)

Note:

- Number of stages depends on the random order of choosing points
- Minimum 1 stage, if first two points chosen are closest
- Maximum $n-2$ stages, if for every point chosen next, there is a change in closest pair

Testing a proposed distance:

- Subdivide the unit square into subsquares, with side $\delta/2$, ie, $N = 2/\delta$ subsquares on each side, N^2 subsquares
- Subsquare S_{st}, for $0 \leq s \leq N-1$ and $0 \leq t \leq N-1$ is

$$S_{st} = \{(x,y) : s(\delta/2) \leq x < (s+1)(\delta/2); t(\delta/2) \leq y < (t+1)(\delta/2)\}$$

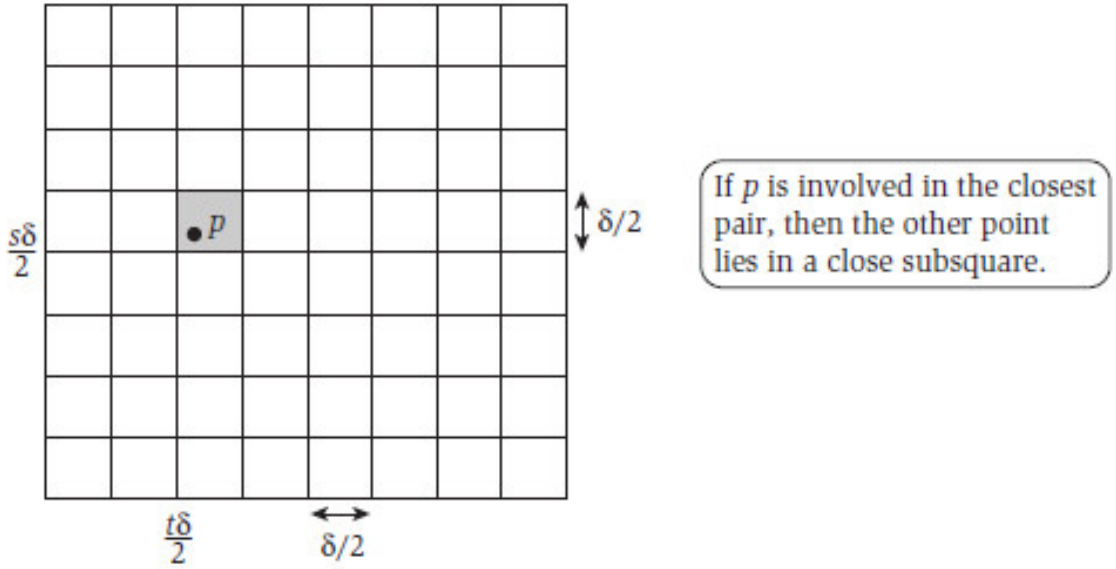
- Observe **first**, that any two points that lie in the same subsquare have distance less than δ.
- **Second**, any two points that are less than δ away from each other must fall in either the same subsquare or in very close subsquares

Designing the algorithm (3): Lemma 1

Lemma 1: If two points p and q belong to the same subsquare S_{st}, then $d(p, q) < \delta$.

Proof:

- If points p and q are in same subsquare, then both coordinates of the two points differ by at most $\delta/2$, and hence
  $d(p, q) \leq \sqrt{(\delta/2)^2 + (\delta/2)^2} = \delta/\sqrt{2} < \delta$



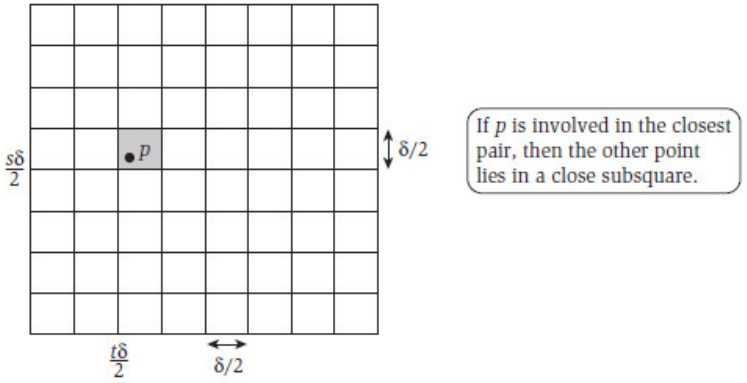
- Subsquares S_{st} and $S_{s't'}$ are close if $|s - s'| \leq 2$ and $|t - t'| \leq 2$.

Designing the algorithm (4): Lemma 2

Lemma 2: If for two points $p, q \in P$ we have $d(p, q) < \delta$, then the subsquares containing them are close.

Proof:

- Consider $p, q \in P$ belongs to subsquares that are not close
- Assume $p \in S_{st}$ and $q \in S_{s't'}$, where one of s, s' or t, t' differs by more than 2. ie, in one of their respective x or y-coordinates, p and q differ by at least δ, thus we cannot have $d(p, q) < \delta$.



- Note: for any S_{st}, the set of close subsquares is a 5×5 grid around it
- At most 25 subsquares need to be checked for new closest pair for a new point p found in S_{st}

Designing the algorithm (5): Summary

- Let $P' \subseteq P$ is set of random points already seen and δ is minimum distance seen so far for P'.
- For each of the points in P', keep track of the subsquare containing it.
- When next point p_i comes, determine which of the subsquare S_{st} it belongs to
- If p_i causes δ to change, there must be some earlier point $p_j \in P'$ at distance less than δ from it
- Thus, p_j must be in one of the 25 squares around S_{st} containing p_i.
- Check each of these 25 squares to see if it contains a point in P' (Lemma 2)
- For each point $p_j \in P'$ in 25 subsquares, compute its distance to current point p_i
- Each of these subsquares contains at most one point of P' (Lemma 1), ie, at most a **constant (25)** number of **distance computations**

Designing the algorithm (6): Summary

- If a new δ' (smaller distance) is found, discard the old subsquares, else **phase** continues
- **If new** δ', update points of $p_j \in P'$ and p_i in newly formed subsquares of new $\delta'/2$ side.
- Continue unless all points are processed

Randomized Closest Pair Algorithm

```
Order the points in a random sequence p_1, p_2, ..., p_n
Let δ denote the minimum distance found so far
Initialize δ = d(p_1, p_2)
Invoke MakeDictionary for storing subsquares of side length δ/2
For i = 1, 2, ..., n:
  Determine the subsquare S_st containing p_i
  Look up the 25 subsquares close to p_i
  Compute the distance from p_i to any points found in these subsquares
  If there is a point p_j (j < i) such that δ' = d(p_j, p_i) < δ then
    Delete the current dictionary
    Invoke MakeDictionary for storing subsquares of side length δ'/2
    For each of the points p_1, p_2, ..., p_i:
      Determine the subsquare of side length δ'/2 that contains it
      Insert this subsquare into the new dictionary
    Endfor
  Else
    Insert p_i into the current dictionary
  Endif
Endfor
```

Dictionary: for storing subsquares

- Need to name a subsquare S_{st} and quickly determine which points of P, if any, are contained in it; thus **Dictionary**
- The universe U of possible elements is set of all subsquares; set S maintained by the data structure are subsquares that contain points from set P' that has been already seen
- For each $p_j \in P'$ keep the subsquare containing it, tagged with index of p_j. Note, $N^2 = (2/\delta)^2$ much larger than n, thus **Hashing**
- **New point** p_i appears; 1) determine S_{st} containing it; 2) perform a **Lookup** for each of the 25 subsquares close to S_{st} for any points pj using **Dictionary** and compute $d(p_i, p')$. if
  - all $d(p_i, p_j) > \delta$: insert p_i in dictionary, ie. map (s, t) of p_i to a hash table slot tagged with p_i
  - any $d(p_i, p_j) < \delta, \delta = d(p_i, p_j)$: discard current **Dictionary**; create new Dictionary **MakeDictionary** with **new** $\delta/2$, ie, insert all points $p' \in P'$ and p_i in corresponding subsquares in dictionary.

Hashing: Randomized implementation of Dictionary

- Hashing is used to maintain a dynamically changing set of elements; Ex: set of news service recording first paragraph of articles
- U: Universe of possible elements / keys, which of very large size; $S \subseteq U$ that needs to be maintained dynamically for insert, delete and search
- Makedictionary: initializes a freash dictionary that stores $S \subseteq U$, initially empty; it supports operations:
  - **Insert(u)**: adds $u \in U$ to S
  - **Delete(u)**: remove u from S
  - **Lookup(u)**: if $u \in S$ returns relevant information of u
- **Example:** Hash function $h(x) = x \mod 250$ maps IP addresses to positions in an array (table) of length 250.

Families of Hash Function (1)

- Hash function must be random but consistent; however, no single hash function behaves well on all sets of data
- Example: if 2^{32} IP addresses to be mapped to 250 slots, atleast $2^{32}/250 \approx 2^{24}$ hashing to same slot (collision), thus **randomization**
- Pick up a hash function at random from some class of functions; this will reduce collisions.
- **Example:** Let number of slots is $n = 257$ (a prime number close to 250). IP address $x = (x_1, x_2, x_3, x_4)$; then $h(x_1, x_2, x_3, x_4) = (87x_1 + 23x_2 + 125x_3 + 4x_4) \mod 257$, where $\{87, 23, 125, 4\} \in \{0, 1, \cdots, n\}$ be any set of 4 coefficients.
- Thus, for any $x = (x_1, x_2, \cdots, x_m)$ and $a = (a_1, a_2, \cdots, a_m) \in \{0, 1, \cdots, n\}$, define $h_a(x) = \sum_{i=1}^{m} a_i x_i \mod n$

Families of Hash function (2)

- Any a_i picked at random $h_a(x)$ is likely to be a good hash function.
- **Property:** if $x = (x_1, x_2, x_3, x_4)$ and $y = (y_1, y_2, y_3, y_4)$ are distinct, and $a = (a_1, a_2, a_3, a_4)$ are chosen uniformy at random from $\{0, 1, \cdots, n-1\}$, then $Pr\left[h_a(x_1, x_2, x_3, x_4) = h_a(y_1, y_2, y_3, y_4)\right] = \frac{1}{n}$
- **Proof:**
  - Let x and y differ by some component $x_4 \neq y_4$
  - Compute $Pr\left[\sum_{i=1}^{4} a_i x_i \mod n = \sum_{i=1}^{4} a_i y_i \mod n\right]$
  - Compute $Pr\left[\sum_{i=1}^{3} a_i(x_i - y_i) \mod n = a_4(y_4 - x_4) \mod n\right]$,
  - After picking up a_1, a_2, a_3 from $\{0, 1, \cdots n-1\}$, a_4, compute probability such that $\sum_{i=1}^{3} a_i(x_i - y_i) \mod n = a_4(y_4 - x_4) \mod n$ holds
  - Let $c = \sum_{i=1}^{3} a_i(x_i - y_i)$, n is prime and $x_4 \neq y_4$
  - Then $a_4 = c(y_4 - x_4)^{-1} \mod n$, probability of this is $\frac{1}{n}$
- Familiy $\mathcal{H}$ of Universal Hash function $\mathcal{H} = \{h_a : a \in \{0, 1, \cdots, n-1\}^m\}$

Application of Universal Hash function

- Very few subsquares S_{st} from the universe U of N^2 subsqares will have points in it.
- Thus, universal hash function $h_a(s,t) = (a_1 s + a_2 t) \mod n$ is used to map subsquare S_{st} to a hash table slot, where p_i is tagged.
- **MakeDictionary()** maps each $p' \in P'$ to the corresponding hash table slot using (s,t), the subsquare S_{st} that p' belongs to

Time complexity (1)

- **Observation 1:** We need atmost n **MakeDictionary**, the worst case, when new δ is found for every point that appears, ie, $O(n)$
- **Observation 2:** a) Algo. maintains closest pair at all times, b)performs atmost $O(n)$ Lookup operations and distance computations (25 distance computed for each new point)
- Let X_i is an indicator $(0-1)$ random variable; ie.

$$X_i = \begin{cases} 1, & \text{if } i^{th} \text{ point causes } \delta \text{ to change} \\ 0, & \text{otherwise} \end{cases} \tag{4}$$

- Let X is a random variable representing the **number of insert operations** performed in the Dictionary
- **Observations 3:** total number of insert operations is $X = n + \sum_{i=1}^{n} i.X_i$
  - the first n: Each point is inserted once when it is first encountered;
  - $\sum_{i=1}^{n} i.X_i$: reinsert all i points when δ has changed due to p_i

Time complexity (2)

Observation 4: For 0-1 random variable X,
$E[X] = 0.Pr[X = 0] + 1.Pr[X = 1] = Pr[X = 1]$
Lemma: $Pr[X_i = 1] \leq \frac{2}{i}$
Proof:

- Let minimum distance among i points $p_1, p_2, \cdots p_i$ be determined by two points p and q
- point p_i can only cause the minumum distance to decrease if $p_i = p$ or $p_i = q$
- Now, the first i points are in random order, thus, any of them is equally likely to be the last chosen point
- the probability that p or q is the last is $\frac{1}{i} + \frac{1}{i} = \frac{2}{i}$

Time complexity (3)

Compute $E(X)$

$$
\begin{aligned}
&= E\left[n + \sum_{i=1}^{n} i.X_i\right] \\
&= n + E\left[\sum_{i=1}^{n} i.X_i\right] \\
&= n + \sum_{i=1}^{n} i.E\left[X_i\right] \\
&= n + \sum_{i=1}^{n} i.\frac{2}{i}, \text{ Lemma and Obs. 4} \\
&\leq 3n = O(n)
\end{aligned}
$$

In expectation, the randomized closest-pair algorithm requires $O(n)$ time plus $O(n)$ dictionary operations.

References

J. Kleinberg and E. Tardos, Algorithm Design,Pearson, Chapter 13.6,13.7

The End